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FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF BIOMEDICAL AND PHARMACEUTICAL TECHNOLOGY

INTRODUCTION TO PHARMACOLOGY

PCT III LECTURE NOTE

FOR

NATIONAL DIPLOMA STUDENTS

(ND I)

BY

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INTRODUCTION

Illness has been man's heritage from the beginning of his existence. Therefore, the search for remedies to combat diseases makes the concept of drugs very important.

DRUG

A drug is any substance used for the purpose of diagnosis, prevention, relief or cure of a disease in man or animals. In other words, a drug is any therapeutic agent, chemical substance or combination of substances that is used or intends to use to prevent or treat ailment/illness or disease.

SOURCES OF DRUGS

1. Natural sources of drugs

Active principles of drugs are mainly present in crude form in minute amounts. These active principles are separated by various techniques or may be used as such in some cases.

(A) **Organic drugs:** Main natural sources of organic drugs are plants, animals and microorganisms.

1. **Plant source:** It is very old and important source of drugs; various parts of the plants are still used for drugs. The following table summarizes some of it.

Part of the plant	Name of the plant	Active Principle	Use
Root	Rauwolfia	Reserpine	Antihypertensive
Bulbs	Urginea	Squill	Emetic, rodenticide
Bark	Cinchona	Quinine	Antimalarial
Rhizome	Ginger	Gingerol	Carminative
Wood	Sandal wood	Sandal wood oil	Urinary antiseptic
Leave	Quassia	Quassin	Stomachic
	Belladonna	Atropine	Anti-muscarinic
	Digitalis	Digitoxin	Cardiac stimulant
Flower	Clove	Eugenol	Local anaesthetic,
	Pyrethrum	Pyrethrin	Insecticide
Fruit	Senna	Senegrin	Purgative
Seed	Nux vomica	Strychnine	Rodenticide
	Physostigma	Physostigmine	Anti-glaucoma
Corn	Colchicum	Colchicine	Anti-gout

These natural occurring chemicals found in plants are called phytochemical, which are found to have biological activity/effect on systems. These are *Tannins, Saponins, Glycosides, Waxes, Steroids, Alkaloids, Anthraquinones, Terpenes, Flavonoids, Carbohydrate, etc.*

2. **Animal sources:** From animal source relatively few but important drugs are obtained. Following are some useful drugs:

Hormones : Insulin, thyroxin, gonadotropins, etc.

Vitamins : Cod liver oil (vit. A).

Vaccines/serra : Anti-rabies vaccine (A.R.V), antitetanic, etc.

Serum : anti-diphtheric serum (A.T.S), etc.

Replacement therapy : Liver extract.

3. **Microorganisms:** Microorganisms are also an important source of drugs. Certain bacteria, fungi and actinomycetes produce antibiotics which forms a major group of therapeutically useful drugs.

Fungi : Penicillin G from *Penicillium* moulds.

Actinomycetes : Streptomycin from *Streptomyces griseus*.

Bacteria : Bacitracin from *Bacillus subtilus*, etc.

(B) **Inorganic:** Drugs obtained from inorganic sources have been simply classified into metals and non-metals. Metalloids which show intermediate properties are usually discussed along with metals. The elements either occur in native state or combined state.

1. **Metals:** Minerals are the main source of metals.

Agent	Use
Magnesium sulphate	Purgative
Calcium carbonate	Astringent

Copper sulphate	Emetic
Ferrous sulphate	Haematinic
Zinc sulphate	Astringent
Bismuth subnitrate	Antiseptic
Lead acetate	Local sedative, antiseptic

2. Non-metals: These are also used for various purposes.

<u>Agent</u>	<u>Use</u>
Iodine (Pot. iodide)	Expectorant
Bromine (Pot. bromide)	Sedative
Hydrogen peroxide	Antiseptic
Sulphur	Insecticide, disinfectant
Carbon (activated charcoal)	Adsorbent

2. Synthetic sources of drugs

Synthetic drugs are prepared in the laboratory with the help of inorganic and organic drugs. Today majority of drugs are obtained synthetically or semi-synthetically. Numerous drugs which were originally obtained from plants are now prepared synthetically. For example, ether and chloroform (Volatile anaesthetics), sulfonamides and quinolones (antimicrobial drugs), paracetamol (analgesic), pentobarbital and thiopental (hypnotic and parenteral anaesthetics), etc.

DRUG NOMENCLATURES

These drugs have nomenclatures that consist of different names applied to one or more ingredients that elicit a pharmacological effect.

- a. **Code number or code designation:** this is a convenient reference for those working with the drug during lab investigation and which is discontinued as soon as the adequate name is given. E.g. Ketamin, a general anesthetic, was given the code number N01AX03. Paracetamol, a synthetic antipyretic, was given the number N02BE01 when it was first synthesized. Chloroquine phosphate, antimalarial drug, was given the code number P01BA01. Also, Mefloquine, a synthetic antimalarial, was given the code number WR142, 490 when it was first synthesized.
- b. **Chemical name:** this is a derived name given according to the rules of nomenclature of a chemical compound. The names describe the chemical structure of a drug of known composition and provide complete and accurate chemical identification. E.g. (RS)-2-(2-Chlorophenyl)-2-(methylamino) cyclohexanone, 4-hydroxyacetanilide, 4(7-chloroquinolin-4-yl)-1,1-diethylpentan-1,4-diamine
- c. **Brand or trade name:** this name designates a formulation and its manufacturer. It is a legal possession of its owner and it is usually selected for its brevity, catchiness and ease of retention and recall. E.g. Ketalar, Panadol, Pfizerquine, Septin, etc.
- d. **Generic, official or common name:** this refers to the established name by which a drug is known as an isolated or synthesized substance irrespective of its manufacturer. E.g. Ketamin, Paracetamol, cotrimoxazole, chloroquine, aspirin, etc.

PHARMACOLOGY

The word pharmacology was derived from a Greek word; "Pharmacōn" meaning drugs and "logos" meaning to study.

Pharmacology is defined as the detailed study of drugs, particularly their actions on living animals, organs or tissues (the action maybe beneficial or harmful). In other words, it can be defined as the study of how chemical agents affect living processes. It is the study of changes brought about in living organisms by chemically acting substances (with the exception of foods), whether used for therapeutic purposes or not. In other words, pharmacology is the study of drugs and chemicals on living system.

Pharmacotherapeutics: is a branch of Pharmacology, which is defined as the study of the therapeutic uses and effects of drugs; this is a study of beneficial and adverse effects of drugs. It also refers to the use of drugs to prevent, treat, and diagnose disease as well as to alter normal functions (such as preventing pregnancy). Pharmacotherapeutics correlates pharmacokinetics and pharmacodynamics with the microbiologic or biochemical aspects of disease. As part of the process of administering drugs to patients, the healthcare's provider must determine what the expected outcome will be. Thus, if an antibiotic for example is prescribed, a reduction in infection is the expected outcome. In order to determine the effectiveness of a drug, the outcome should be measurable in some way: through observation (decreased inflammation), patient reports (less pain), and/or assessment (decreased WBC count).

Even though two patients may have the same diagnosis, this does not necessarily mean that the same treatment or medication is indicated. When assessing a patient's

need for medications and choosing the best medication for the patient, the healthcare provider must consider many issues:

- The condition for which the patient is to be treated.
- The patient's current drugs (including prescribed, over-the counter medications, and illicit drugs).
- Possible drug interactions that may occur.
- Contraindications.
- Patient's mental status and ability to comply with directions.
- Need for follow up testing (such as periodic blood tests).
- Costs of the drug and whether or not the patient has insurance to cover these costs.
- Patient's social history (homeless status, presence or absence of support system, access to transportation or other services).
- Pregnancy or breastfeeding status.
- The route of administration (oral, rectal, parenteral).
- Absorption, distribution, metabolism, and excretion of drug.
- Adverse effects of drugs.
- Current medical conditions that may affect drug activity (such as kidney or liver disease).

Based on the assessment of the patient, the following pharmacotherapeutic steps are indicated:

- ✓ Define the patient's problem for which treatment is indicated.
- ✓ Outline the therapeutic objective.
- ✓ Choose the appropriate treatment based on current practice and evidence-based research.
- ✓ Verify the suitability of the chosen treatment.
- ✓ Begin treatment.

- ✓ Provide information to the patient and caregivers about the drug, instructions for use, and any precautions.
- ✓ Carry out ongoing assessment and continue, modify, or stop treatment.

Pharmacotherapeutics encompasses many different types of therapy, some of which may overlap:

- **Acute therapy:** This type of therapy is indicated with acute, sudden and/or critical illnesses, such as pneumonia or cancer. Acute therapy usually needs to begin immediately if the patient is to improve or recover. Treatment is often more intense and may involve multiple drugs and ongoing assessment. Examples of acute therapy include epinephrine for anaphylaxis and chemotherapy for cancer.
- **Chronic/Long-term/Maintenance:** This type of therapy is intended for maintenance and to prevent a patient's condition from worsening or to control symptoms but not generally expected to bring about a cure. Patients with chronic disease, such as diabetes (insulin), bipolar disorder (lithium), and heart disease (diuretic, beta-blockers), may need to take medications for many years and often for the rest of their lives. Maintenance therapy may also be used to replace other drugs, such as long-term treatment with methadone to reduce heroin use. Long-term therapy may also be indicated for non-illness-related reasons, such as oral contraceptives to prevent pregnancy and hormone therapy for transgender individuals.
- **Replacement therapy:** This type of therapy serves to replace something that is missing in the body and often critical, such as insulin. Some replacement therapy is needed throughout the patient's life, but others, such as hormone replacement therapy, may be used for a period of months or years to relieve symptoms. Replacement therapy may also be used to treat deficiencies of vitamins and minerals, such as vitamin D replacement therapy and iron for iron deficiency anemia.

• **Palliative therapy:** This type of therapy is intended to provide comfort rather than cure. Palliative therapy is often used in the late stages of disease, such as cancer and heart disease. Palliative therapies include:

- i. Analgesics (such as ibuprofen and opioids) to relieve pain and dyspnea. Anti-emetics (such as metoclopramide) to relieve nausea and vomiting.
- ii. Benzodiazepines (such as diazepam and lorazepam) to relieve anxiety.
- iii. Stool softeners (such as docusate sodium) to relieve anxiety.
- Antidepressants (such as amitriptyline and fluoxetine) to relieve constipation
- iv. Antidiarrheals (such as loperamide) to relieve depression.
- v. Steroids (such as dexamethasone) to relieve diarrhea
- vi. Antimuscarinic agents (such as hyoscine butylbromide) to reduce respiratory secretions.

• **Supportive:** This type of therapy is intended to maintain body functions while the patient recovers from disease or injury. This may include the administration of packed red bloods for a patient with blood loss, electrolytes to provide the electrical energy needed for many body functions, and IV fluids for dehydration.

• **Prophylactic therapy:** This therapy is intended to prevent disease rather than treat or cure existing disease. Prophylaxis may include peri-surgical administration of antibiotics, rabies injection after exposure to a potentially rabid animal, routine vaccinations, and antiretroviral medications after exposure to HIV.

• **Empiric therapy:** This therapy is administered based on probabilities that a disease exists in the absence of confirming data. For example, if a patient has signs of *Clostridium difficile* infection, immediate treatment may begin before the culture and sensitivities can confirm the diagnosis. Empiric therapy is often given based on symptoms when patients are in critical condition and a diagnosis has not yet been formulated.

Pharmacovigilance: WHO defines pharmacovigilance as the science and activities relating to the collection, detection, assessment, monitoring and prevention of adverse effects or any other possible drug-related problem. It focuses on investigating and monitoring adverse drug reactions after medicinal products are licensed.

Aim and scope of pharmacovigilance

- To improve patient care and safety in relation to medicines and all medical and para-medical interventions.
- To improve public health and safety in relation to the use of medicines.
- To contribute to the assessment of benefit, harm, effectiveness and risk of medicines.
- To promote understanding, clinical training and effective communication to health professionals and the public.

Toxicology (Paracelsus 1500AD, the founder of modern toxicology): is also a branch of pharmacology which is the study of mechanisms by which drugs and chemicals in the environment produce unwanted effects. In other words, is the qualitative and quantitative study of the adverse or toxic effect of chemicals and other anthropogenic materials or xenobiotics on living organisms. Modern toxicology focuses on the adverse effects of toxic substances (including toxins, venoms, and toxicants) at molecular level.

Common poisons and their antidotes

Acetaminophen	Acetylcysteine
Organophosphate (e.g. insecticides)	Atropine
Tricyclic antidepressants, quinidine	Sodium bicarbonates
Calcium channel blockers	Intravenous calcium
Iron salts	Deferoxamine
Digoxin and other cardiac glycosides	Digoxin antibodies
Ethylene glycol, methanol	Ethanol (same as alcohol used for drinking), given intravenously
Benzodiazepines	Flumazenil
Beta blockers	Glucagon
Opiates, opioids drugs	Naloxone
Carbon monoxide (by inhalation)	Oxygen (at high concentration), known as bariatric therapy

PHASES OF CLINICAL TESTING

Phase I: first administration to healthy human volunteers, following animal studies; to determine pharmacokinetics and pharmacodynamics.

Phase II: First administration to patients, small studies; to determine therapeutic efficacy, dose range, pharmacokinetics and pharmacodynamics.

Phase III: Double blind, large sample; to compare a new drug to placebo or standard of care, safety and efficacy.

Phase IV: Post-marketing surveillance, wide distribution; to determine rear adverse reactions, effect of long-term use and to determine ideal dosing.

ROUTE OF DRUG ADMINISTRATION AND SITE OF ACTION

Drugs are contained in dosing units called formulations or dosage forms. A route of administration is a way of getting a drug onto or into the body.

Before a drug is administered, a route of administration as well as a suitable dose and dosage form (e.g. tablet, capsule, injection, solution, syrup, suspension, suppository, etc.) maybe selected in order to ensure that the drug will reach its site of action in pharmacologically effective concentration.

Common routes of drug administration

Route	Advantages	Disadvantages
Enteral routes		
1. Oral (Po)	➤ This is a common and easiest route. ➤ It is a safe route since in over dose it can be managed. ➤ Example, oral antibiotics; ciprofloxacin, levofloxacin, doxycycline, erythromycin, cephalexin, cloxacillin, amoxicillin etc.	➤ Some drugs are destroyed by gastro-intestinal juice as Penicillin. ➤ Some drugs are not absorbed from GIT as gentamycine. ➤ Some drugs cause local irritation to the mucous membrane. ➤ Irregularity of absorption by food or anticholinergic drugs.
2. Buccal or Sublingual (SL)	➤ No first-pass metabolism as Glycerol trinitrate. ➤ To avoid delay in the rate of absorption.	➤ Drugs are carried directly to the liver by portal vein & may be inactivated by first-pass metabolism.

3. Rectal (PR)	➤ Example loratadine, mirtazapine, nitroglycerin, etc. ➤ Is use when a drug causes irritation by other routes or when patient cannot swallow drug. ➤ Used for local and systemic effects. ➤ Example, prednisolone, hydrocortisone, mesalazine, etc.	➤ Irritation to the mucous membrane of the mouth. ➤ Excessive salivation promotes swallowing. ➤ Not for most drugs or drugs with high doses. ➤ It causes discomfort to some patients. ➤ Drug may migrate to different position. ➤ Absorption maybe erratic.
<i>Parenteral routes</i> 1. Intravenous (IV)	➤ Large volume of preparation can be given, with control the rate of administration. ➤ Rapid onset of drug action, so, it is preferred in emergencies. ➤ Drugs that are rapidly destroyed can be infused continuously.	➤ Increase chance for adverse drug reaction. ➤ I.V fluids are aqueous solutions only, suspension cannot be given to avoid embolism. ➤ Thrombus is liable to occur especially with prolonged infusion. ➤ It is not safe route thus the drug must be given slowly.

2. Intramuscular (IM)	➤ Administration of drugs that cannot be absorbed by the gut or irritant to be given by other route as: Anti cancer drugs. ➤ Gentamicin, cisplatin, mifamfugin, morphine, dopamine, etc. ➤ It is a common route and easier to inject than intravenous injection. ➤ More rapid than subcutaneous route and less affected by peripheral circulatory failure. ➤ Larger volume maybe used compared to subcutaneous solutions. ➤ Site of injection is the muscle of arm, thigh, gluteal region. ➤ Example, penicillin G, vaccines, testosterone, etc.	➤ Irritating drugs maybe painful ➤ Different rates of absorption depend upon muscle group injection and blood flow.
3. Subcutaneous (SC)	➤ Generally used for insulin injection. ➤ The volume of injected fluid is usually 1ml or less. ➤ Example, insulin, opioids.	➤ Only aqueous solutions are used because aqueous suspensions and oleaginous suspensions and solutions are irritant and cause pain. ➤ Poor absorption in case of peripheral circulatory failure.

4. Intraperitoneal (IP)	➤ A relatively large volume of non-irritant drugs is given. ➤ The rate of absorption is faster than subcutaneous and Intramuscular route. ➤ The site of injection is the lower part of the abdomen, the needle should not touch the intestine but only in the peritoneal cavity. ➤ Example, melphalan, mitomycin C, cisplatin, etc.	➤ It is given by the professionals.
<i>Other routes</i> 1. Transdermal	➤ Used for lipid-soluble drugs with low dose and low molecular weight. ➤ The volume of preparation injected is rarely more than 0.2 ml because the tissue volume is small and compact. ➤ The isotonicity is very important because the route is mostly for diagnostic purposes.	➤ The absorption is slow due to poor circulation. ➤ Permeability of skin varies with condition, anatomic site, age and gender. ➤ Non-isotonic solutions may cause false signs, which give false positive results.

2. Inhalation	➤ Examples, nicotine, scopolamine, estrogen and testosterone; etc. ➤ Maybe used for local and systemic effects. ➤ Rapid action when used for systemic administration like inhalational anesthesia. ➤ Direct action (topical) as (Salbutamol) for asthma with less systemic side effect. ➤ Example, fluticasone, beclomethasone, mometasone, etc.	➤ It needs a special apparatus. ➤ The substance should be not irritant. ➤ The patient must be conscious.
3. Topical	➤ Easy to administer. ➤ This route provides a high local concentration of the drug without affecting the general circulation. ➤ Example, corticosteroids, antifungals, antivirals, antibiotics, antiseptics, etc.	➤ Effects are mainly limited to site of application. ➤ Absorption may occur when there is tissue damage.

Factors governing choice of route of drug administration

The choice of appropriate route in a given situation depends both on drug as well as patient related factors. Mostly common sense considerations, feasibility and convenience dictate the route to be used.

1. Physical and chemical properties of the drug (solid, liquid, gas, solubility, stability, pH, irritancy).
2. Site of desired action-localized and approachable or generalized and not approachable.
3. Rate and extent of absorption of the drug from different routes.
4. Effect of digestive juices and first pass metabolism on the drug.
5. Rapidity with which the response is desired (routine treatment or emergency).
6. Accuracy of dosage required (IV and inhalational can provide fine tuning).
7. Condition of the patient (unconscious, vomiting).

BROAD DIVISIONS OF PHARMACOLOGY

Pharmacology is broadly divided into:

1. Pharmacokinetics
2. Pharmacodynamics

1. PHARMACOKINETICS

Pharmacokinetics is the study of rate process associated with drug absorption, distribution, metabolism and excretion of drug; *it tells us how the body handles drugs*. It also deals with quantitating drug and or metabolites following administration until completely removed from the body.

It involves changes in drug concentration in the different body fluids and tissues.

Pharmacokinetics examines the movement of a drug over time through the body. Understanding pharmacokinetic principles, combined with specific information

regarding an individual drug and patient, underlies the individualized optimal use of the drug (e.g. choice of drug, route of administration, dose and dosing interval).

1. Absorption

It is defined as the movement of the drug from the site of administration into the plasma. It also concerns the processes of entry of a drug into the systemic circulation from the site of its administration. The determinants of absorption are those described for drug permeation. *For solid dosage forms, absorption first requires dissolution of the tablet or capsule.* This liberates the drug to be absorbed into the local circulation from which it will distribute to its sites of action.

Intravascular administration (e.g., IV) does not involve absorption, and there is no loss of drug, bioavailability 100%. With extravascular administration (e.g. [PO; oral], intramuscular [IM], subcutaneous [SC], inhalation), less than 100% of a dose may reach the systemic circulation because of variations in bioavailability.

Factors affecting the rate and extent of drug absorption

1. *Partition coefficient of the drug ($P_{oil/water}$):* This is the relative solubility of a drug in oil (lipid) and water. Drugs with high lipid solubility can rapidly diffuse across cell membranes. E.g. *anesthetics* are very lipid soluble and therefore have a rapid action of onset.
2. *Blood flow:* This depends on the nature of local blood flow at the site of administration. E.g. sublingual vessels provide significant blood flow and therefore, rapid absorption.
3. *Molecular size:* Drugs with small molecular weight absorb faster than high molecular drugs.
4. *pH and drug ionization:* Drugs are weak acids and weak bases, therefore, they have an ionized and non-ionized form. pH and pK_a determined the ratio of ionized : non-ionized. Non-ionized forms cross cell membranes much faster than ionized forms.

5. *Total surface area*: The small intestine has villi, which increase the surface area for absorption; hence, the small intestine is the primary site of absorption for most oral drugs.

Bioavailability

Bioavailability indicates the extent to which a dose of drug reaches its site of action. In other words, is measure of the fraction of drug that reaches the systemic circulation in an unchanged state following administration.

Factors affecting bioavailability

1. Drug absorption
2. Metabolism in the gut wall
3. Hepatic first-pass effect

IV and Intra-arterial dose has 100% bioavailability (bioavailability = 1). E.g. penicillin G is destroyed by gastric enzymes and needs to be administered IV.

Hepatic first-pass effect

The metabolism of a drug by the liver, following administration, but before it reaches systemic circulation, and can exert an effect. With oral administration, drugs are absorbed into the portal circulation and initially distributed to the liver. For some drugs, their rapid hepatic metabolism decreases bioavailability.

GI tract (absorption) → portal vein → liver (first-pass metabolism) → systemic circulation.

Example of drugs with a high hepatic first-pass effect include: lidocaine, propranolol, morphine, etc.

Example of drugs with a low hepatic first-pass effect include: diazepam, digoxin, phenytoin, warfarin, etc.

2. Distribution

After absorption or systemic administration into the bloodstream, a drug distributes into interstitial and intracellular fluids depending on the particular physicochemical properties of the drug. *Is a process by which drugs move between body compartments to the site of action.* The rate of delivery and amount of drug distributed into tissues is determined by; cardiac output, regional blood flow, capillary permeability and tissue volume. Initially, liver, kidney, brain, and other well-perfused organs receive most of the drugs, whereas delivery to muscle, most viscera, skin, and fat is slower. Tissue distribution is determined by the partitioning of drug between blood and the particular tissue.

Factors affecting the rate and extent of drug distribution

1. Physicochemical properties of the drug (e.g. partition coefficient)
2. pH of fluid
3. plasma protein binding
4. cardiac output
5. regional blood flow
6. percentage body fat

3. Metabolism (Biotransformation)

Metabolism is the process of a conversion of one chemical species to another chemical specie. *The biotransformation of drugs and into more hydrophilic metabolites is essential for their elimination from the body, as well as for termination of their biological and pharmacological activity.* In general, biotransformation reactions generate more polar, inactive metabolites that are readily excreted from the body. However, in some cases, metabolites with potent biological activity or toxic properties are generated.

Drug metabolism or biotransformation reactions are classified as either:

Phase 1: functionalization reactions (oxidation, reduction, or hydrolytic reactions)

Phase 2: biosynthetic (conjugation) reactions (glucuronidation, sulfation, acetylation and methylation).

The enzyme systems involved in the biotransformation of drugs are localized primarily in the liver (cytochrome p450 (CYP) enzymes and enzymes for conjugations), although every tissue examined has some metabolic activity. Interactions at the level of drug metabolism form the basis of many drug interactions. Some drugs can inhibit Cytochrome P450 (CYP) independently of being substrates. For example, the common antifungal agent, ketoconazole is a potent inhibitor of CYP3A4 and other CYPs. Co-administration of ketoconazole with the anti-HIV viral protease inhibitors, Saquinavir reduces the clearance of Saquinavir and increases its plasma concentration and the risk of toxicity.

A drug may induce its own metabolism (enzyme inducers). One potential consequence of this is a decrease in plasma drug concentration resulting in loss of efficacy. A drug may also induce metabolism of other drugs resulting in low concentration of the second drugs. Omeprazole, Phenobarbital, Rifampins are enzyme inducers and reduce concentration of many drugs when co-administered.

Factors affecting drug metabolism

1. Genetic polymorphism of metabolizing enzymes: individual may metabolize drugs faster or slower depending on their genotype.
2. Enzymes inhibition due to competition from other drugs and micronutrient.
3. Enzymes induction: certain medications enhance gene transcription leading to an increase in the activity of a metabolizing enzyme.
4. Liver dysfunction cause by disease: hepatitis, alcoholic liver disease, biliary cirrhosis and hepatocellular carcinoma may decrease metabolism; however,

due to the liver's reserve, the reduction in metabolism may not be clinically significant.

5. Renal disease:

- *Age*: neonates and the elderly have reduced ability to biotransform, therefore adjust dosing accordingly.
- *Nutrition*: vitamin and mineral deficiencies may also affect metabolizing enzymes.
- *Alcohol*: chronic alcohol consumption can induce the P450 enzymes and increase the generation of the toxic metabolites and hepatocellular damage.
- *Smoking*: cigarette smoke induces CYP1A2 thereby increasing the metabolism of some drugs.

4. Excretion

Excretion is defined as the irreversible loss of a drug in a chemically unchanged or unaltered form. In other word, it is the removal of drug from the body. *It is the final step in the ADME process and consists of a series of pathways that remove an administered drug and/or its metabolites from the body.* Excreted drugs are either eliminated in their original, unmetabolized form, or they can be eliminated following metabolic biotransformation.

The two principal organs responsible for drug elimination are the kidney and the liver. The kidney is the primary site for removal of a drug in a chemically unaltered or unchanged form (i.e. excretion) as well as for metabolites. The liver is the primary organ where drug metabolism occurs. The *lungs*, occasionally, may be an important route of elimination for substances of high vapor pressure (i.e. gaseous anesthetics, alcohol, etc.). Another potential route of drug removal is a *mother's milk*. Although

not a significant route for elimination of a drug for the mother, the drug may be consumed in sufficient quantity to affect the infant.

Factors affecting drug elimination

1. Physicochemical properties of drug: e.g. molecular size
2. Blood flow to the kidney
3. Alteration of urine pH and diuresis (alkaline diuresis for treating salicylates and Phenobarbital intoxication).
4. Protein binding of the drug: limits glomerular filtration (GF) but doesn't limit active tubular secretion.
5. Biological factors:
 - Sex: renal excretion is 10% lower in female than in male
 - Age: the renal excretion in newborn is 30-40% less in comparison to adults.
 - Old age: glomerular filtration rate (GFR) is reduced and tubular a function is altered which results in slow excretion of drugs and prolonged half-life.
 - Disease state: kidney, liver and renal dysfunction, and uremia